

QuickU

Demand Forecasting & Stock Planning August 2022

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01

Introduction



Business Overview

10 Categories

Consisting of: Vegetables & Fruits, Milk & Dairy Products, Cooking Ingredients & Seasonings, Cooking Kits, Snacks, Frozen Meat, Beverages, Frozen Foods, Protein (Chicken & Eggs), Staple Foods

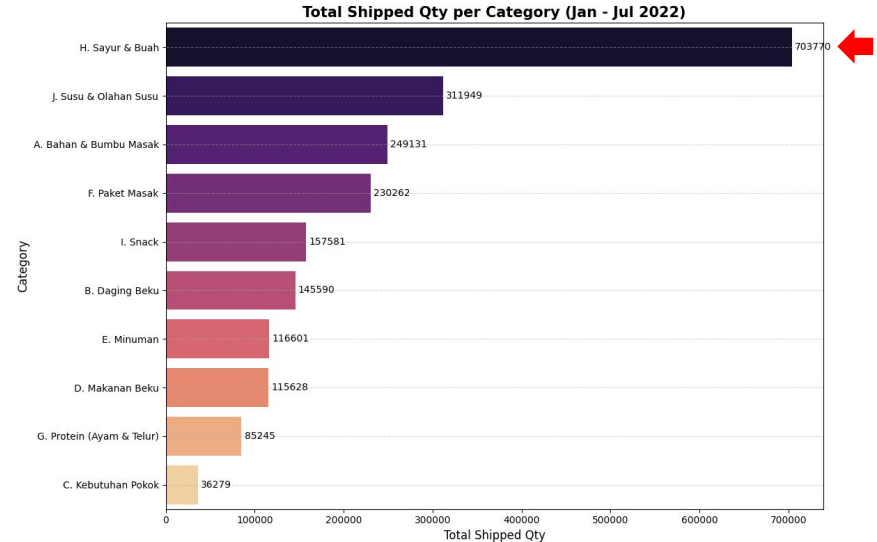
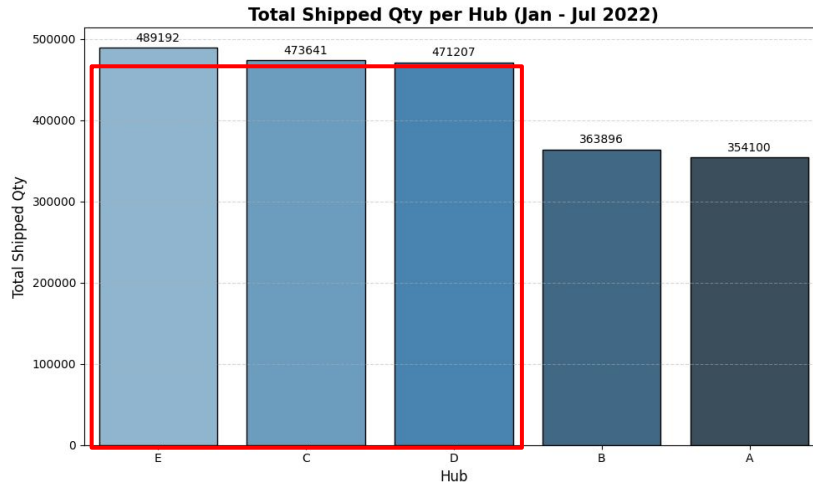
5 Hub (Locations)

28 Weeks

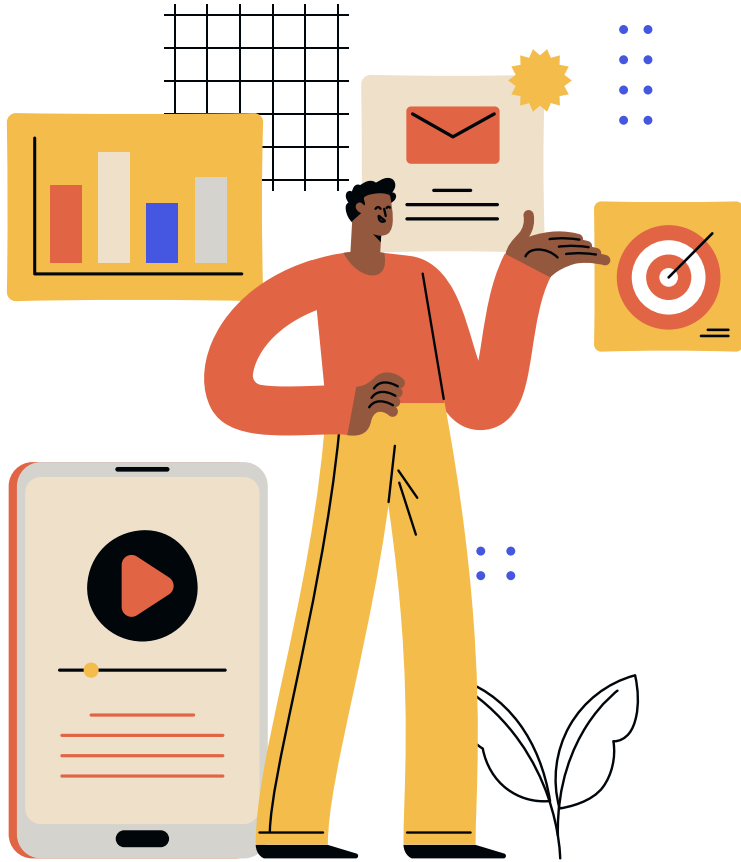
Total weeks over 7 months



Business Overview



- The **highest demand** is in the **fresh product** category, especially **vegetables and fruits**.
- **Hubs E, C, and D** are **core operational hubs** that support most of the demand.
- The business model seems to be **more focused** on fast-moving **fresh goods** than **basic necessities**.



02

Business Question



Business Question

Business Problems

Determine the **most accurate forecasting method** and predict **demand for August 2022**, using available historical data, with completion prior to August stock preparation.

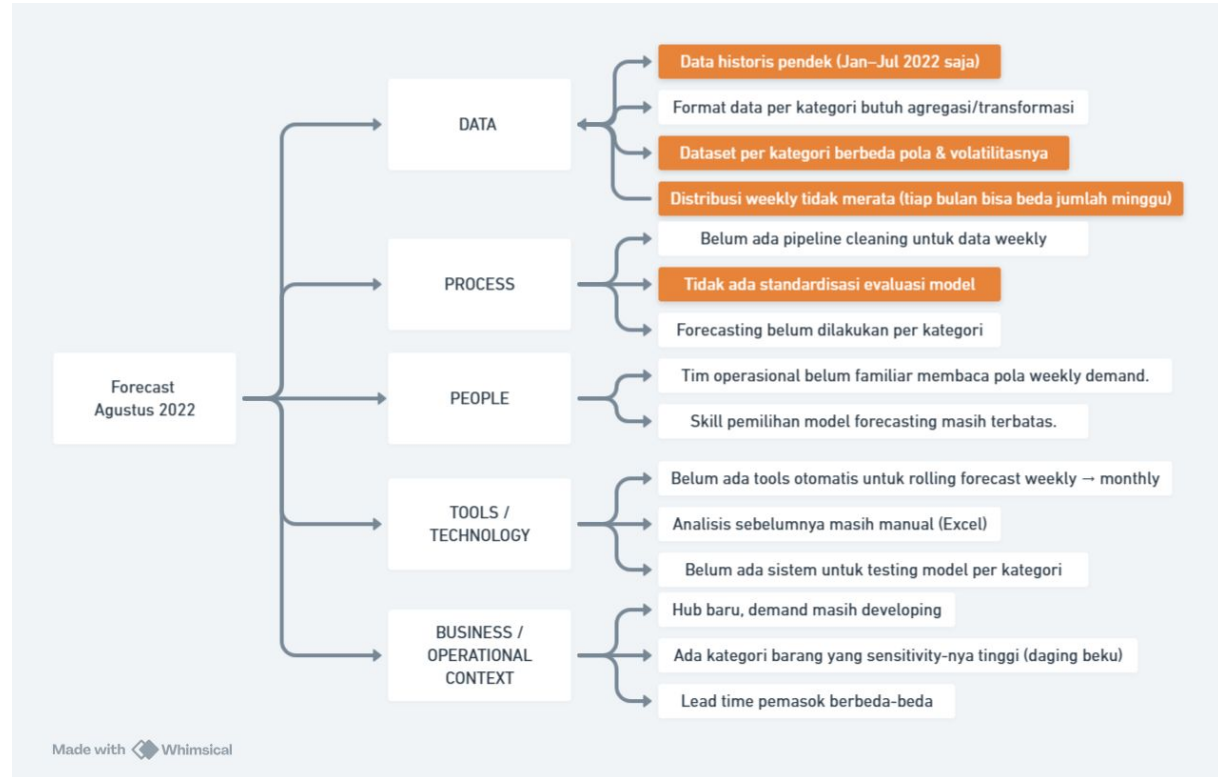
Project Scope

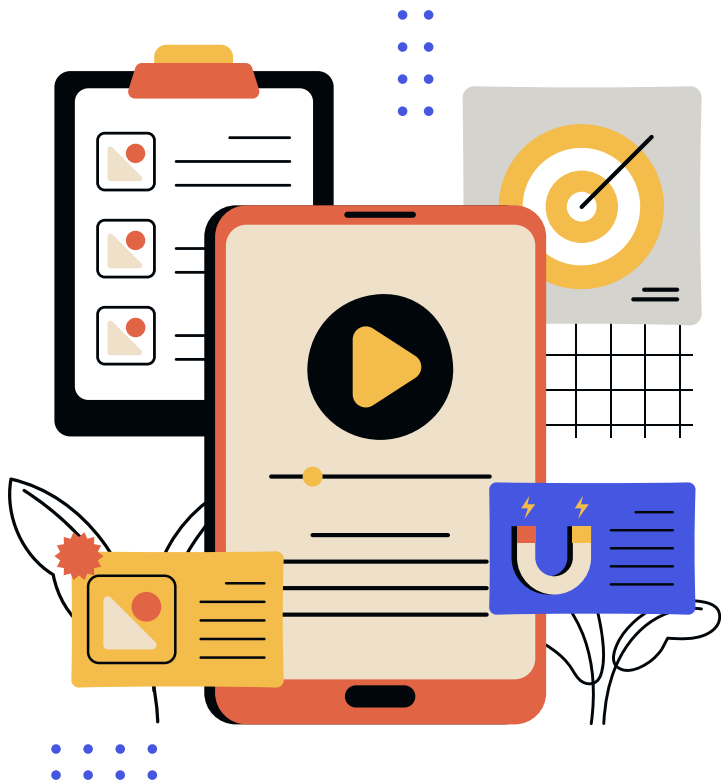
predict **demand for August 2022** from **historical data januari - july 2022**

Objectives

1. August 2022 demand forecast for the new hub using the available historical dataset.
2. Select the method with the lowest error.
3. Complete the analysis before the stock preparation process begins

Root Cause Analysis

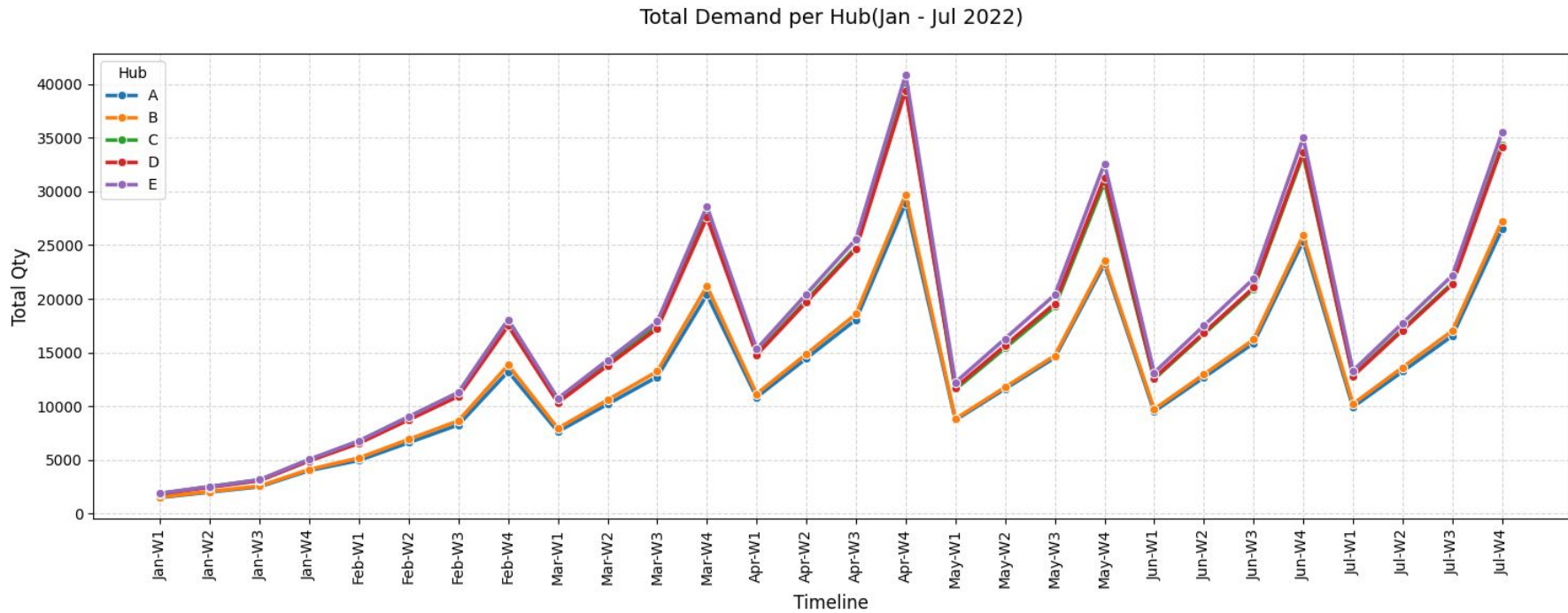




03

Analysis

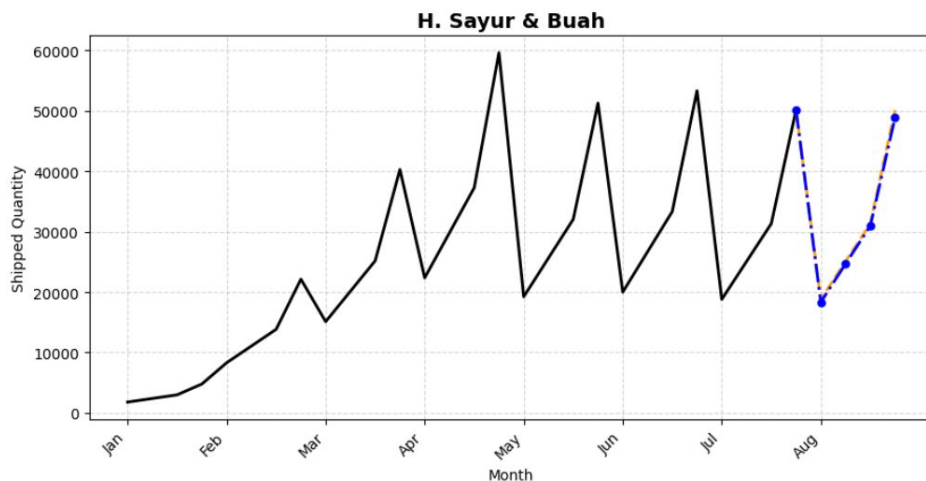
EDA & Forecast
Python



Model Multiplicative



EDA - Vegetables & Fruits



Model Forecast Comparison

Sensasional Naive
125297

Holt-Winters
123409

MAPE 6.38%

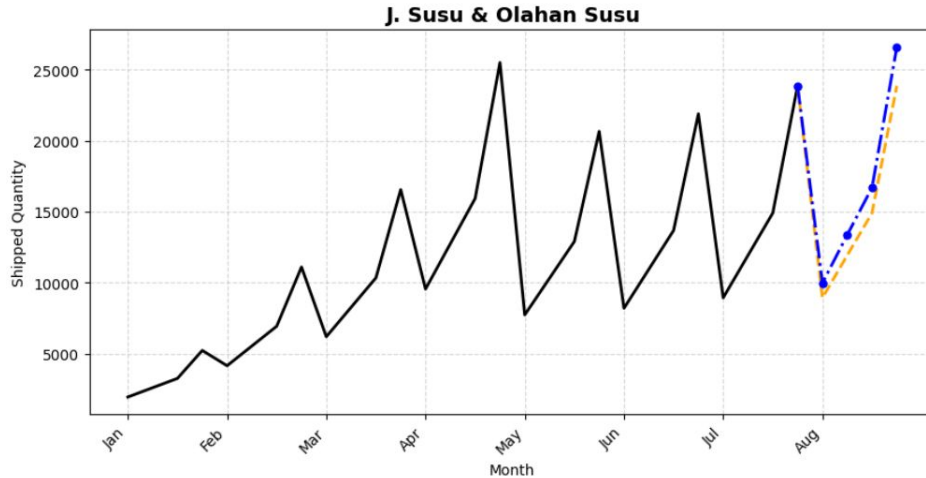
MAPE 18.94%

Accuracy 93.62%

Accuracy 81.06%



EDA - Milk & Dairy Products



Model Forecast Comparison

Sensasional Naive
59641

Holt-Winters
67108

MAPE 8.25%

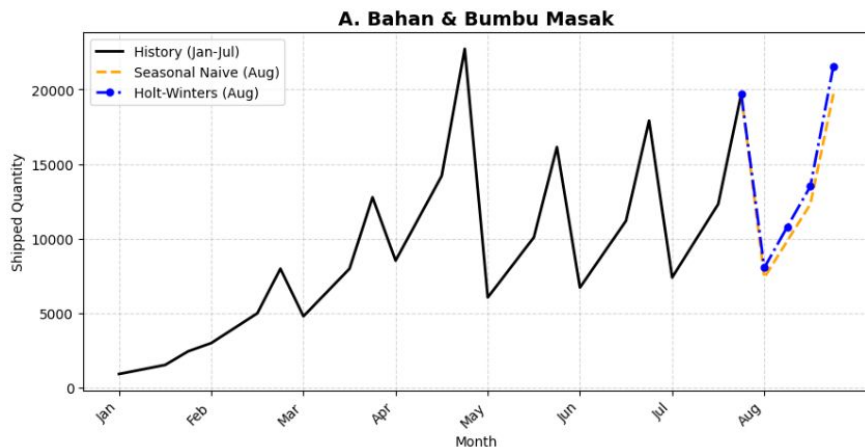
MAPE 3.69%

Accuracy 91.75%

Accuracy 96.31%



EDA - Cooking Ingredients & Seasonings



Model Forecast Comparison

Sensasional Naive
49246

Holt-Winters
59547

MAPE 9.08%

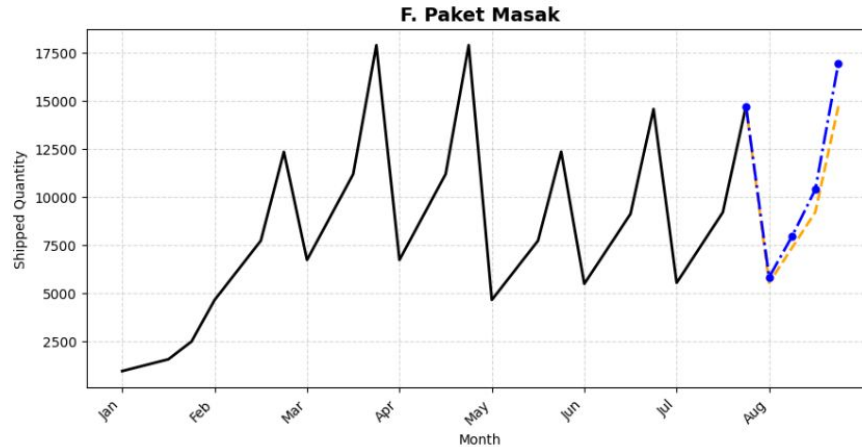
MAPE 5.49%

Accuracy 90.92%

Accuracy 94.51%



EDA - Cooking Kits



Model Forecast Comparison

Sensasional Naive
36740

Holt-Winters
42636

MAPE 0.99%

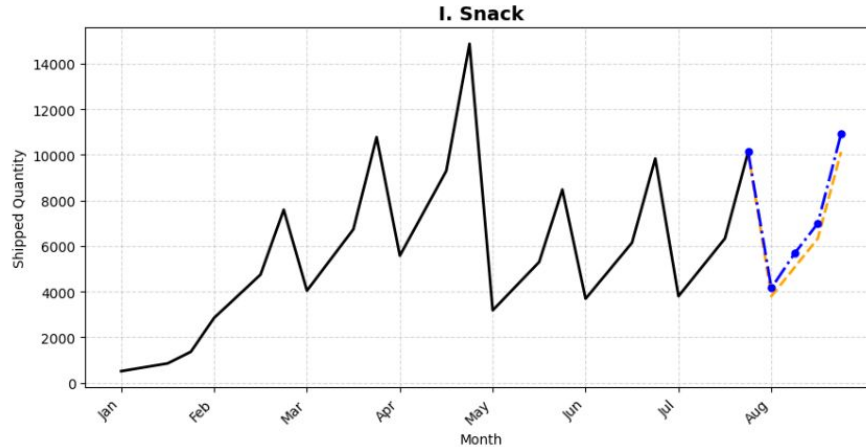
MAPE 7.43%

Accuracy 99.01%

Accuracy 92.57%



EDA - Snacks



Model Forecast Comparison

Sensasional Naive
25319

Holt-Winters
29206

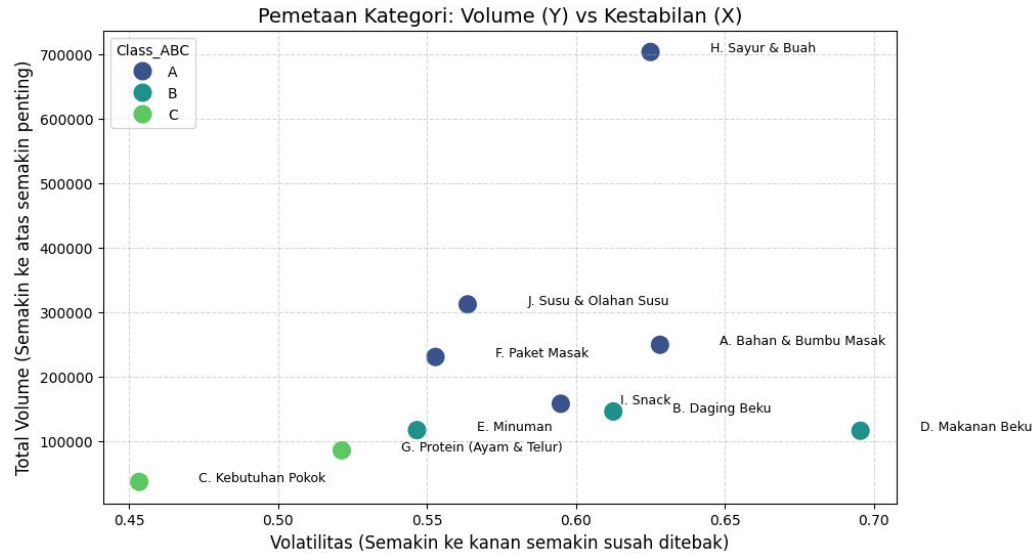
MAPE 2.91%

MAPE 5.56%

Accuracy 97.09%

Accuracy 94.44%

Clustering



Matrix

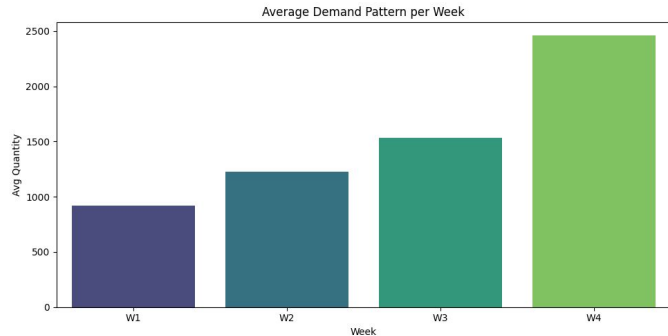
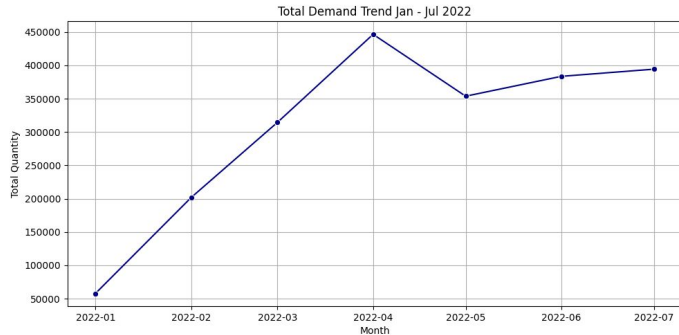
AX (Hot Items + Stable)	Vegetables & Fruits, Milk & Dairy Products, Cooking Ingredients & Seasonings, Cooking Kits, Snacks
BZ (Safety Stock)	Frozen Meat, Beverages, Frozen Foods
CY (Small Seasonal)	Protein (Chicken & Eggs)
CZ (Problem Child)	Staple Foods

04

Result & Recommendation



Observation



Business Growth

Demand trends are rising consistently.



Weekly Pattern

Weeks 1 & 2 steady demand and Week 4 demand skyrockets 2-3 times



Category Variations

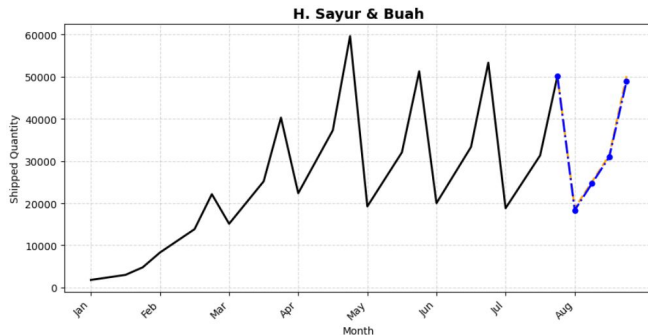
Large categories have stabilized over the past two months and Small/medium categories, despite low volumes, are still trending upward.

Background



Business Phase

Our main category (Vegetables & Spices) has entered the Maturity phase, so the pattern is stable and repetitive



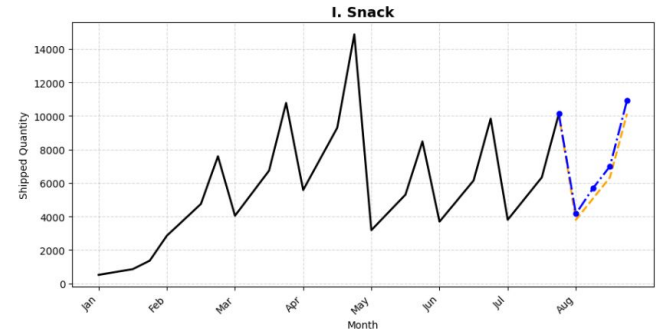
Short-Term Data

Historical data from the past seven months shows that consumer behavior in stable categories tends to repeat the patterns seen in the previous month.



Growth Phase

Complementary categories (c) are still in the Introduction/Growth phase, which is why they are highly volatile and trending upward.



Insight

Holt-Winters Too Aggressive

for the stable category (Class A),
Holt-Winters tends to be overconfident
and prone to overstocking.

Seasonal Naive is the winner

the simple “Copy-Paste Last Month”
method has been proven to have
fewer errors.

Hybrid is Key

Stable goods require a conservative (Naive) method.
Growth goods require an aggressive (Holt-Winters)
method.

Recommendation

Procurement Strategy - CRITICAL



Category AX (Vegetables/Spices)

Use **Seasonal Naive forecast** figures.
Do not buy excessively following false trends. Focus on **maintaining availability**, not increasing stock recklessly.

“Play it safe.”



Safety Stock

Add an **extra 15-20% reserve stock** specifically for categories ending in 'Z' (Frozen Meat & Staples) because the error rate is still high.



BY/CY Category (Frozen/Snacks)

Use **Holt-Winters forecast** figures.
We must **dare to stock** more because demand is on the rise.

“Go for it.”

Recommendation



Warehouse Strategy (Manpower)

W4 Full Team: Maintain the ban on leave during the fourth week of August. Although category A is stable, the volume remains enormous and requires a lot of manpower.

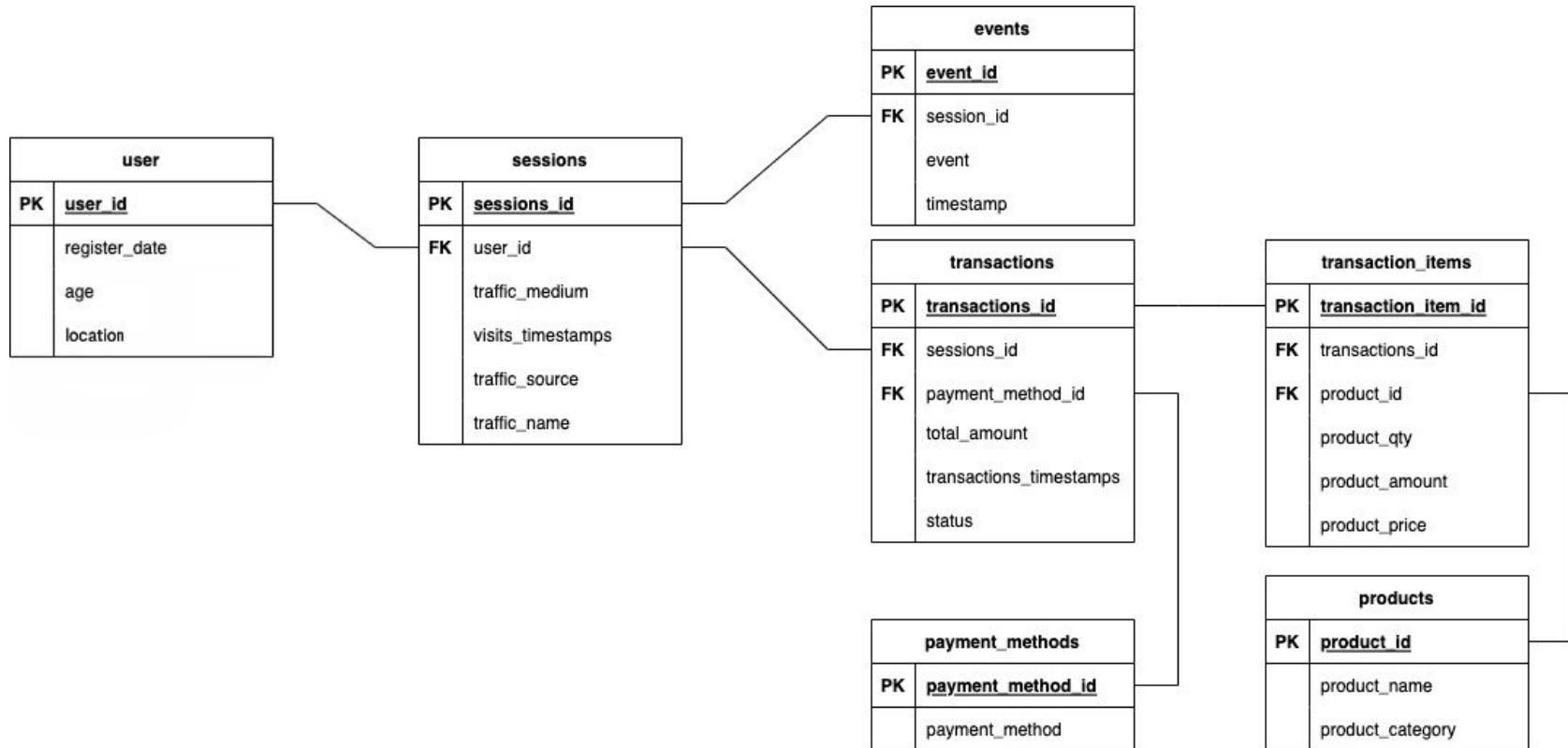


Evaluation (Monitoring)

Weekly Review: Check Category A stock every Friday. If Week 1 turns out to be busier than the Naive prediction, immediately switch to the Holt-Winters strategy for the rest of the month (dynamic in nature).

APPENDIX

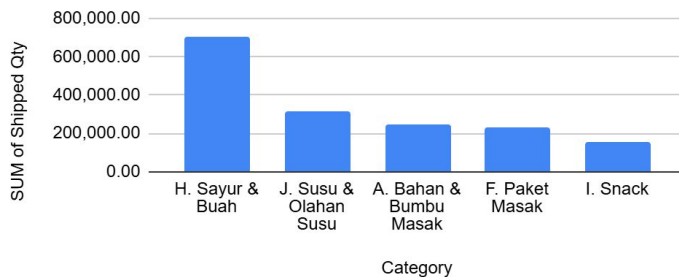
ERD



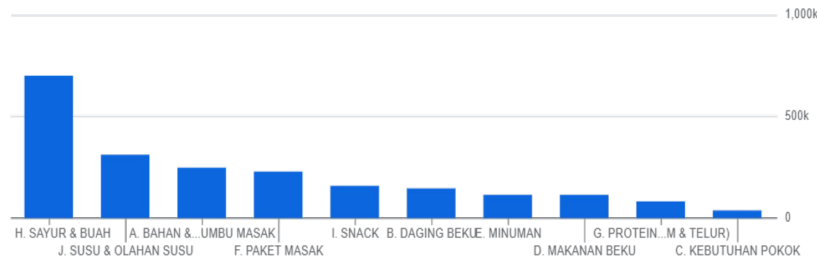
Dataset Overview

1. Time range 7 Months : Jan - Jul 2022 With **1.400 rows** of weekly data from **5 different hubs**.
2. Total volume (Shipped Qty) **2.152.036** unit.
3. Data Quality: 100% **Clean**. **No missing** values, so it is ready for immediate processing.
4. Locations (Hub) : 5 Hubs (A, B, C, D, and E).
5. Products : 10 Categories (ranging from Cooking Ingredients & Spices to Milk & Dairy Products).
6. Granularity data is recorded **weekly** (W1-W4), but we **aggregate it monthly** for restocking accuracy.

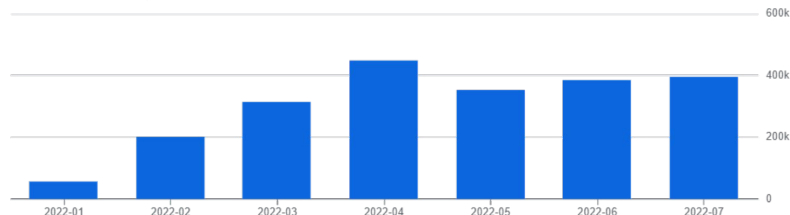
Total Qty per Kategori



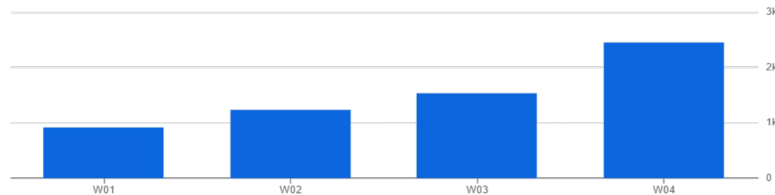
total_demand by category



total_demand by month



avg_weekly_demand by week



Result & Recommendations



Business Growth

Demand trends are rising consistently.



Payday Effect

Extreme spike in Week 4

Holt-Winters (Multiplicative)

Highest accuracy (captures upward trends and salary spikes)



Stock Priority

Critical (AX/AY): Vegetables & Fruits, Protein, Cooking Packages (Large Volume + Clear Pattern).

Volatile (BZ/CZ): Frozen Meat, Basic Necessities (Medium/Small Volume + Unpredictable Pattern).

THANK YOU



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